

as detecting recurrence, participants required further evidence before using HPV status, but a majority would consider its use in other aspects such as detecting location of a primary. Currently there is a low level of modification of treatment for HPV positive patients; however many participants are willing to consider it in light of further evidence. European participants are statistically more likely to rate modification of treatment as more important compared to USA ($p=0.003$) and Canadian ($p=0.028$) participants. However they adopt a more cautious approach in using HPV status to detect disease recurrence or when monitoring for metastatic disease.

Conclusions: This is the first study of its kind investigating the variation in terms of HPV testing and perception of its importance on an international level. These results suggest that a standardised testing method is required in order to compare clinical results from different countries, and greater awareness of clinical trials in this area is required. The treatment of head and neck squamous cell carcinoma in relation to HPV status varied depending on country and the size of the medical institution. Therefore there is a need for a common policy in testing and use of HPV information in order to allow for standardisation for head and neck squamous cell carcinoma patients across all countries and institutions. The current level of testing and willingness of clinicians to participate in this area of research recognises their interest in their area. Therefore - potential exists for the initiation of large, multi-national trials with the aim of reducing toxicity for these patients who appear to have a better prognosis with the current standard of care.

PO-0949

AUDIT OF RUTHENIUM-106 BRACHYTHERAPY FOR POSTERIOR UVEAL MELANOMA IN THE SCOTTISH OPHTHALMIC ONCOLOGY SERVICE

D. Ritchie¹, M. Gregory², T. Betts², S. Drummond², W. Kincaid³, F. Roberts⁴, E. Kemp², P. Cauchi²

¹Beaumont West of Scotland Cancer Centre, Clinical Oncology, Glasgow, United Kingdom

²Gartnavel Hospital, Tennant Institute of Ophthalmology, Glasgow, United Kingdom

³Gartnavel Hospital, Radiologist, Glasgow, United Kingdom

⁴Gartnavel Hospital, Pathologist, Glasgow, United Kingdom

Purpose/Objective: 1) To establish the outcome of treatment with ruthenium plaque radiotherapy for choroidal melanoma less than 6mm depth with regards to visual acuity, tumour response, eye retention, local recurrence, metastases, all-cause mortality and melanoma-related mortality. 2) To correlate outcome regarding tumour response to radiation dose and tumour margins.

Materials and Methods: A retrospective case note review of patients with uveal melanoma treated consecutively between November 1995 and December 2008 was performed. Those receiving Ruthenium-106 brachytherapy combined with local resection were excluded.

Results: During this period, 308 patients were treated with ruthenium plaque. 68 cases were excluded due to prior tumour treatment, adjuvant diode laser or local resection. 240 cases were analysed. Median age was 65 years (range:29-94), gender distribution was equal. 14 ciliary body and 226 choroidal melanomas were treated. Pretreatment median tumour thickness was 3.1mm (0-8), median largest basal diameter was 9.1mm (1.5-20.9). During a median follow up of 47 months, initial response to brachytherapy was observed in 97%. Median tumour thickness and largest basal diameter decreased to 1.9mm (0-9.8) and 7.5mm (0-18.7) respectively at final follow up. Final tumour response rate was 80.3%. Additional treatment was required for recurrence/non-response ($n=39$), or other complications ($n=13$). Complications (17.5%) included scleral necrosis, neovascular glaucoma or raised IOP, vitreous haemorrhage, non-proliferative radiation retinopathy, macular oedema, optic neuritis and chronic uveitis. The 5-year predicted eye retention rate was 85.6%. Vision of $\geq 6/18$ was retained in 58.5%. The 5-year predicted overall survival and disease-specific survival was 82.7% (95% CI:77.2%, 88.2%) and 92.5% (95% CI:88.4%, 96.6%) respectively. 50 % of tumour recurrence/non response occurred in patients with lesion size more than 10.9mm, representing 25% of the study population. A detailed analysis of dose received and tumour margins will be presented.

Conclusions: Ruthenium-106 brachytherapy was associated with high rate of ocular retention, preservation of vision, low tumour-related mortality. A 2mm margin from tumour edge to plaque edge is associated with a lower risk of local tumour relapse.

PO-0950

CAN AN ELEVATION OF C-TELOPEPTIDES IN CANCER PATIENTS PREDICT BONE DISSEMINATION?

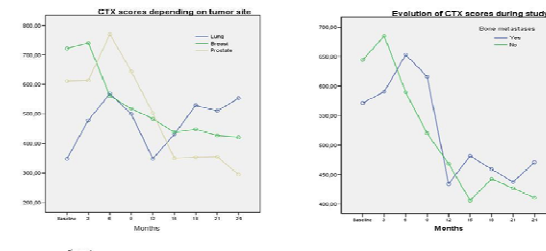
C. López Carrizosa¹, J. Zapatero Ortuño¹, P. Samper Ots¹, V. Jerviz Guía¹, M.L. Cuoselo Paniagua¹, M. Domínguez Morilla¹, J. Saez Garrido¹, M. Martín de Miguel¹

¹Hospital Central de la Defensa Gómez Ulla, Radiotherapy Oncology, Madrid, Spain

Purpose/Objective: Early diagnosis of bone metastasis (BM) is not possible with current imaging techniques. An elevation of C-Telopeptides (CTX) could support early diagnosis and treatment of BM. Our objective is to analyze the evolution of CTX and Alkaline Phosphatase (AP) in patients with breast, prostate and lung cancer and its predictive value for early diagnosis of BM.

Materials and Methods: Prospective study carried out with 21 patients in cancer stages II-III treated with radiotherapy were included. Minimum follow-up of 1 year. Mean age 60 years (35-85). 14 patients were women, 7 were men. By tumor sites: 14 (66.7%) breast, 4 (19%) prostate and 3 (14.3%) lung. Laboratory tests including hemogram, biochemical determination, serum CTX and urinary phospho-calcium metabolism were performed at baseline and every three months as well as a bone gammagram at baseline and upon the diagnosis of BM and after 24 months. Statistical analysis was carried out using SPSS v15.0, considering $p<0.05$ statistically significant.

Results: Of the initial 21 patients, 3 abandoned the study. Of the remaining 18, 15 are currently free of disease and three have developed BM. Of these 3, 2 are breast cancer patients who developed BM 9 months after beginning the study, the other being a lung cancer patient who developed BM within 6 months of beginning our study. All three patients received further treatment with zoledronic acid, chemotherapy or hormone therapy. At the end of the study, both breast cancer patients are controlled while the lung cancer patient presents progression of the primary tumor. Baseline values of CTX: 589.95 ± 256.81 pg/ml ($106.80 - 1440$ pg/ml), baseline values of AP: 77.86 ± 17.19 U/L. By genders, baseline levels are higher in women than in men ($p=0.0001$). By tumor sites (Fig 1), after 3-6 months, CTX levels are higher in prostate and breast cancer than in lung cancer; after 9-12 months CTX levels gradually increase in lung cancer until achieving higher levels than the other sites by the end of the study (p not significant). CTX levels are higher in patients with BM vs. patients without BM (p , not significant) and in tumor progression vs. tumor control (p , ns). AP shows a gradual increase after 6-9 months in patients with BM ($p=0.020$) (Figura 1).



Conclusions: Baseline CTX scores are higher in women, probably due to secondary osteoporosis or natural or induced menopause.

A gradual increase in CTX scores in lung cancer patients, higher than prostate or breast cancer, was observed after 12-24 months.

AP and CTX scores are higher in patients with BM than in those without BM.

CTX could be a predictor for bone dissemination, though a higher patient recruitment would be necessary to establish a relationship.

POSTER: RTT TRACK: PATIENT CARE

PO-0951

A RETROSPECTIVE ANALYSIS OF AN OROPHARYNGEAL CANCER DATABASE, WITH FOCUS ON A HPV SUBGROUP, IN ONE CENTRE IN IRELAND

M. Dixon¹, T. O'Dwyer², A. Craig¹

¹Trinity College Dublin, Discipline of Radiation Therapy, Dublin, Ireland Republic of

²University College Dublin, Mater University Hospital, Dublin, Ireland Republic of